

The Perron–Frobenius Rank-One Step in Cirac–Perez-Garcia–Schuch–Verstraete 2017

Appendix C.2

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1 The assertion

Appendix C.2 of Cirac–Pérez-García–Schuch–Verstraete relates the strong area law and zero correlation length for a simple matrix product density operator to an outer-product factorization of a finite trace matrix [CPGSV16, Appendix C.2, Lemmas C.3–C.4].¹

The paper constructs positive neighboring operators $\eta_{k,h}$ and defines

$$T_{k,h} = \text{tr}(\eta_{k,h}).$$

Thus T is entrywise nonnegative. Under the strong area law the source argues that T is primitive, in the sense that some power of T has all entries strictly positive. Under zero correlation length it obtains

$$\text{tr}(T^N) = \text{tr}(T)$$

for every positive integer N , with $\text{tr}(T) = 1$ after normalization. The proof then invokes fixed-point theory for primitive nonnegative matrices and concludes that T has one eigenvalue equal to 1 and all other eigenvalues equal to 0. From this it concludes the desired factorization

$$T_{k,h} = a_k b_h.$$

The last implication is the point at issue. Primitivity and the equalities of positive-power traces control the nonzero eigenvalues of T , but they do

¹For traceability, this note corresponds to issue #1041. The underlying question was isolated in issue #832; the finite counterexample was added in pull request #849; and the positive-semidefinite replacement theorem was added in pull request #917. The source passage is `Papers/1606.00608/MPD0-22-12-17-2.tex:1394--1499`.

not exclude nilpotent Jordan structure on the generalized eigenspace for the eigenvalue 0. Such nilpotent structure is invisible to traces of positive powers and can prevent T from having rank one.

2 The counterexample

Consider the real 3×3 matrix

$$T = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{6} & \frac{1}{6} & \frac{2}{3} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{pmatrix}.$$

All entries are nonnegative, and

$$T^2 = P := \frac{1}{3} \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}.$$

Hence T is primitive, since T^2 has all entries strictly positive. Also $P^2 = P$ and $PT = TP = P$, so $T^N = P$ for every $N \geq 2$. Since $\text{tr}(T) = \text{tr}(P) = 1$, it follows that

$$\text{tr}(T^N) = \text{tr}(T) = 1$$

for every positive integer N .

Nevertheless T is not an outer product. If $T = ab^\top$, then the entry $T_{1,1} = 1/2$ implies $a_1 \neq 0$. The entry $T_{1,3} = 0$ then forces $b_3 = 0$, but this contradicts $T_{2,3} = 2/3$. Thus the universal matrix implication

$$\text{primitive} + \text{constant positive-power traces} \implies \text{rank one}$$

is false for finite entrywise nonnegative real matrices.²

This example has the form $T = P + \frac{1}{6}xy^\top$, where $x = (1, -1, 0)^\top$ and $y = (1, 1, -2)^\top$. The perturbation satisfies $(xy^\top)^2 = 0$ and is annihilated by P on both sides. It therefore leaves all traces of positive powers unchanged from the Perron projector while raising the rank of T .

²The formal counterexample is `TNLean/Archive/PerronFrobeniusRankOneCounterexample.lean`, especially the matrix definition at lines 63–65 and the theorem `counterexample` at lines 150–154. It is also summarized in `docs/counterexamples.md`: 19--27.

3 The corrected statement

The formal development proves a corrected sufficient matrix theorem. Let T be a finite real square matrix. If T is positive semidefinite, $\text{tr}(T) = 1$, and $\text{tr}(T^N) = \text{tr}(T)$ for every positive integer N , then T has an outer-product factorization $T = ab^\top$.³ Primitivity is not needed for this corrected matrix theorem once positive semidefiniteness and trace normalization are assumed.

The proof is spectral. Positive semidefiniteness makes T Hermitian and diagonalizable with nonnegative real eigenvalues λ_i . The trace normalization gives $\sum_i \lambda_i = 1$. The trace-power hypothesis at $N = 2$ gives $\sum_i \lambda_i^2 = 1$. For nonnegative numbers with sum one, this forces exactly one λ_i to be 1 and all the others to be 0. Therefore the diagonal form has rank one, and unitary conjugation gives an outer-product factorization of T .

This corrected theorem does not refute the intended MPDO result. It identifies a matrix-level hypothesis sufficient for the rank-one step. The remaining question is whether the trace matrix $T_{k,h} = \text{tr}(\eta_{k,h})$ arising from the neighboring operators is Hermitian or positive semidefinite. Positivity of each $\eta_{k,h}$ gives $T_{k,h} = \text{tr}(\eta_{k,h}) \geq 0$ entry by entry, but entrywise nonnegativity is not matrix-level positive semidefiniteness or Hermitian symmetry of T .⁴

4 Conclusion

The proof in Appendix C.2 contains an unjustified matrix step as written: a primitive entrywise nonnegative matrix with constant traces of positive powers need not have rank one. The obstruction is a nilpotent Jordan component at the zero eigenvalue, which is not detected by the trace identities.

The formal development proves a corrected sufficient theorem: positive semidefiniteness, trace normalization, and the same trace-power hypothesis imply the required rank-one factorization. What remains for the unconditional MPDO argument is not another Perron–Frobenius statement for arbitrary nonnegative matrices, but an MPDO-specific proof that the

³The formal statements are `Matrix.PosSemidef.trace_powers_constant_implies_rank_one` and `Matrix.primitive_trace_powers_constant_implies_rank_one_of_posSemidef` in `TNLean/Algebra/PerronFrobenius/RankOne.lean:181--240`.

⁴The entrywise nonnegativity result is `MPOTensor.ExplicitEtaOperators.traceMatrixRe_nonneg` in `TNLean/MPS/MPDO/SimpleLocalStructure.lean:306--318`. The positive-semidefinite reformulation for the rank-one conclusion is `MPOTensor.sal_zcl_implies_rank_one.T_of_posSemidef` in `TNLean/MPS/MPDO/SimpleLocalStructure.lean:350--366`. The corresponding blueprint discussion is `blueprint/src/chapter/ch02b_mpdo.tex:568--680`.

trace matrix obtained from the η -operators has the Hermitian or positive-semidefinite structure needed by the corrected theorem. Thus the source proof has a genuine unsupported intermediate assertion, while the intended MPDO theorem itself has not been disproved.

References

- [CPGSV16] J. Ignacio Cirac, David Pérez-García, Norbert Schuch, and Frank Verstraete. Matrix product density operators: Renormalization fixed points and boundary theories. Preprint; published version: *Annals of Physics* 378 (2017), 100–149, 2016.